

Discussion 3

Wednesday, April 15, 2026

Complete this worksheet in **a group of 3-5 students**. When you submit, use the **groups feature** on Canvas to add all the students in your group, and submit jointly.

On the next two pages of this worksheet are one question from the practice exam posted on Canvas. A fake student, Drew Blank, has taken the practice exam and gotten every question incorrect. Drew filled in their answers, and then wrote their reasoning underneath each question.

Underneath their reasoning for each part, there is open space for you to write in. Read each of Drew's responses, correct them to the correct answer, explain why Drew's reasoning was incorrect, and explain the reasoning behind the correct answer.

Name: Drew Blank Student Email: dblank @uci.edu

1. The Youth Risk Behavior Survey (YRBS) is a survey conducted by the Centers for Disease Control and Prevention (CDC) in the spring once every two years, to learn about health-related risky behaviors in adolescents in the U.S. The most recent available data comes from the 2023 study. In 2023, the CDC randomly sampled classes at a nationally representative sample of public and private U.S. high schools (grades 9-12: 14-18 year old students). The survey was completed anonymously via a questionnaire booklet.
- (a) What were the observational units in this study?
- ☐ the students who completed the survey
 - ☐ the classes that were selected to participate in the survey
 - ☒ the questionnaire booklets given to the students to complete
 - ☐ the risky behaviors the students report doing

Student reasoning: The observational units are the questionnaire booklets because those booklets are what the researchers will ultimately be observing when they receive them after the survey is done.

Correction:

- (b) What was the population of interest in this study?

students who took the survey

Student reasoning: The population of interest is the group of students who took the survey because we are interested in learning about their risky behaviors, which is why we are giving them the survey.

Correction:

- (c) The sampling design was two-stage. In the first stage, schools in the U.S. were randomly selected for the study. What type of sampling is this?
- ☐ simple random sampling
 - ☐ stratified sampling
 - ☐ cluster sampling
 - ☒ systematic sampling

Student reasoning: The type of sampling in this stage was stratified sampling because the researchers grouped the students by school, the strata, and then sampled those.

Correction:

- (d) In the second stage of sampling, entire classes were randomly selected from within each school. What type of sampling is this?
- ☒ simple random sampling
 - ☐ stratified sampling

- ☐ cluster sampling
- ☐ systematic sampling

Student reasoning: The type of sampling in this stage was simple random sampling because the classes were selected completely at random from the school.

Correction:

- (e) What choice made by the CDC helped mitigate response bias?

sampling design (probability sampling)

Student reasoning: The CDC used probability sampling so that all currently-enrolled students have a fixed probability of being in the sample. Then, there shouldn't be a particular group of people systematically missing from the sample.

Correction:

- (f) One of the questions asked about marijuana use. 29.5% of respondents said that they had used marijuana at least once in their life. Which of the following is true?

- ☐ 29.5% is a statistic.
- ☒ 29.5% of all adolescents in the U.S. have used marijuana at least once in their life.
- ☐ If the survey were repeated with a different group of students, the value of 29.5% would remain the same.
- ☐ None of the above statements are true.

Student reasoning: 29.5% of students in the sample said they have used marijuana at least once, and since this sample is representative, we can say 29.5% of all adolescents have used marijuana at least once.

Correction:

- (g) Which study design most closely matches the design of the YRBS?

- ☒ cohort study
- ☐ case-control study
- ☐ cross-sectional study
- ☐ matched pairs experiment
- ☐ completely randomized experiment

Student reasoning: The study is a cohort study because the CDC surveyed a group of students (a cohort) about their health-related risky behaviors.

Correction: